

// AI BLOG FACTORY — TECH GUIDE SERIES

Digital guides for students, creators & tech learners

THE PROMPT ENGINEERING PLAYBOOK

How to get exactly what you want from AI —
every single time. No theory. Just systems.

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FROM THE AUTHOR

I have spent hundreds of hours talking to AI. Most of those conversations were frustrating. The answers were vague, off-topic, or just wrong. Then I learned one thing that changed everything: the prompt is everything. This book is everything I have learned — condensed into the shortest useful form. No filler. No motivation speeches. Just patterns that work.

// TABLE OF CONTENTS

01	The Anatomy of a Bad Prompt	03
02	The C.O.R.E. Framework	05
03	Context Engineering	08
04	Role-Playing and Personas	10
05	Chain-of-Thought and Reasoning	13
06	Iterative Refinement Loops	15
07	Subject-Specific Prompt Systems	17
08	The Master Prompt Library	20
09	Quick-Reference Cheat Sheet	24

The Anatomy of Bad Prompts

Before we build better prompts, we need to understand why most fail. This is not about AI being dumb. It is about humans being imprecise.

THE THREE FAILURE MODES

Every bad prompt falls into one of three categories:

- > Vague — "Help me with my assignment." (Which assignment? What topic? What format?)
- > Overloaded — "Write a 500-word essay about machine learning, include three examples, cite sources, and make it sound professional." (Too many competing instructions.)
- > Missing Context — "Explain p-values." (Explain to whom? A statistician? A first-year student? A business executive?)

THE ONE-LINE FIX

Before sending any prompt, ask yourself:

"Who is answering? What do they need to know? What should the answer look like?"

If you cannot answer all three, your prompt is not ready.

THE SPECIFICITY LADDER

Prompts exist on a spectrum from vague to precise. The higher you climb, the better your results:

```
Level 1: "Help me with Python"
Level 2: "Help me with Python lists"
Level 3: "How do I remove duplicates from a Python list?"
Level 4: "I am a beginner. Show me 3 ways to remove
         duplicates from a Python list, with code
         examples and a comparison of speed."
```

Most students prompt at Level 1 or 2. Top performers live at Level 4.

The C.O.R.E. Framework

Four elements. Every great prompt contains all four. Most bad prompts contain one — or none.

C – CONTEXT

Who you are, what you are working on, and what you already know. AI needs to calibrate its answer to YOUR level.

```
BAD: "Explain neural networks"
GOOD: "I am a first-year Data Science student.
       I understand basic Python and high-school math.
       Explain neural networks at my level."
```

O – OBJECTIVE

What do you want the AI to actually produce? "Explain" is different from "compare" is different from "debug." Specify the verb.

R – RESPONSE FORMAT

Do you want a paragraph? A bulleted list? Code? A table? Tell the AI. It will follow your format instruction precisely.

E – EXAMPLES

This is the secret weapon. Show the AI one example of what a good answer looks like. It will match that pattern.



CORE PROMPT TEMPLATE

Context: "I am [background] working on [task]. I know [knowledge]."

Objective: "I need [specific thing]."

Response: "Give me [format]."

Example: "Here is an example of what I want: [sample]."

Save this template. Use it for your next 20 prompts. Compare the results to your old prompts.

Context Engineering

Context is the single biggest factor in prompt quality. This chapter makes you dangerously good at providing it.

THE PROGRESSIVE DISCLOSURE METHOD

Instead of dumping all context into one prompt, build it progressively across three rounds:

- > Round 1: State the problem and your level.
- > Round 2: Add constraints, format preferences, and examples.
- > Round 3: Refine based on the AI response — add corrections and ask for a rewrite.

THE "EXPLAIN IT LIKE" SHORTCUT

Fastest context-setting technique in this book:

"Explain [topic] LIKE I AM:"

- a 15-year-old → simple analogies, no jargon
- a first-year student → foundational, with examples
- a professional → advanced, assume knowledge
- a teacher → complete, with common mistakes noted"

CALIBRATION PROMPTS

Start important sessions with one of these:

- > "Before you respond, ask me: (1) What is my level? (2) What format do I want? Then give your best answer based on my replies."
- > "Act as if you are my personal tutor who has known me for a semester. I learn best with examples and analogies."

IMPORTANT

Do not lie about your level. If you say "I am an expert" and the AI assumes you understand advanced concepts, the answer may confuse you — not help you.

Role-Playing and Personas

This technique transforms generic AI into a specialist. The same model gives radically different answers depending on who it pretends to be.

THE TOP 10 PERSONAS FOR STUDENTS

HOW TO ACTIVATE A PERSONA

```
"From now on, act as a [PERSONA].  
Your answers should be [TRAIT].  
I am a [YOUR LEVEL] working on [TASK]."
```



MULTI-PERSONA TECHNIQUE

For complex topics, ask the same question to two personas.
Then ask: "Compare these two answers. Which is more accurate and why?"
This gives you a built-in quality check.

Chain-of-Thought and Reasoning

This is the most powerful technique in this chapter. It forces AI to slow down and think step by step — producing answers that are dramatically more accurate.

HOW IT WORKS

AI models sometimes jump to conclusions. Chain-of-thought forces them to show their work — like a student showing their math working. The result: fewer mistakes, better explanations.

TRIGGERS THAT ACTIVATE REASONING

- > "Think step by step." — the classic, always works
- > "Show your reasoning before the answer."
- > "Walk me through your thinking as if I know nothing."
- > "First, identify the key concepts. Then explain each one. Then give the answer."

WHEN NOT TO USE IT

Chain-of-thought adds length. Do not use it for:

- > Simple factual questions — "What is the capital of France?"
- > Formatting tasks — "Convert this to a table."
- > Short summaries — "Summarize this in 3 sentences."

OVERUSE WARNING

Adding "think step by step" to every prompt wastes tokens and time.

Use it for: math, logic, code debugging, complex analysis.

Skip it for: simple facts, format requests, short tasks.

Iterative Refinement Loops

The best AI outputs come from conversation — not a single prompt. This chapter teaches you the art of the follow-up.

THE THREE-STEP LOOP

Step 1: Draft — Get a rough answer with a basic prompt.

Step 2: Refine — Point out what is wrong, missing, or needs to be different.

Step 3: Polish — Ask for the final version incorporating all feedback.

REFINEMENT PHRASES THAT WORK

```
"That is close, but [SPECIFIC ISSUE]. Fix that."  
"The second paragraph is perfect. Apply that  
quality to the rest."  
"I need this in [DIFFERENT FORMAT]."  
"Add a section about [MISSING TOPIC]."  
"Make it more [CONCISE / DETAILED / TECHNICAL]."  
"The tone is too formal. Rewrite as if explaining  
to a friend."
```

MULTI-PASS COMPLEX TASKS

For research papers, reports, or complex assignments, do not try to get everything in one pass. Use the chapter-by-chapter approach.



THE GOLDEN RULE

Never accept the first answer for anything important.

The first draft is always imperfect. Your job is to iterate.

Most students stop at draft one. Top performers iterate 3-5 times.

Each iteration costs 30 seconds. Each one improves quality noticeably.

Subject-Specific Prompt Systems

Different subjects need different prompt strategies. This chapter gives you ready-made systems for the most common student subjects.

MATHEMATICS AND STATISTICS

```
"I am working on [PROBLEM TYPE].  
Show me the solution step by step.  
After each step, explain WHY we do it,  
not just WHAT we do.  
Use [ANALOGY TYPE] if possible."
```

PROGRAMMING AND DEBUGGING

```
"I am getting this error: [PASTE FULL ERROR]  
Here is my code: [PASTE RELEVANT SECTION]  
I expected: [EXPECTED BEHAVIOR]  
what I got: [ACTUAL BEHAVIOR]  
Explain what went wrong in simple terms."
```

ESSAY WRITING AND RESEARCH

```
"I am writing about [TOPIC].  
I need a thesis statement, 3 supporting  
arguments with sources, and a counterargument.  
My audience: [WHO IS READING]  
My level: [YOUR LEVEL]."
```

PRESENTATIONS AND SLIDES

"Create a [NUMBER]-slide outline for [TOPIC].
Each slide should have: title, 3 bullet points,
speaker notes, and a suggested visual.
Audience: [WHO], Tone: [FORMAL/CASUAL]."

CROSS-SUBJECT INSIGHT

Notice the pattern: every subject prompt includes CONTEXT
(who you are), OBJECTIVE (what you want), and FORMAT
(how it should look). The C.O.R.E. framework applies everywhere.

The Master Prompt Library

Twenty prompts you can copy, paste, and use today. Organized by category.

STUDY AND LEARNING (5 PROMPTS)

1. "Explain [topic] like I am 15. Give 3 real examples."
2. "What are the 5 most testable concepts in [subject]?"
3. "Quiz me on 10 questions about [topic]. Grade me."
4. "I got [WRONG ANSWER]. Why is that wrong?
What misconception caused it?"
5. "Create a study schedule for [exam] on [date]."

CODING (5 PROMPTS)

6. "Build [project] in [language] with comments."
7. "Debug: [paste error + code]. Explain simply."
8. "Optimize this code. Show before and after."
9. "Write tests for: [paste function]."
10. "Explain this code line by line: [paste]."

WRITING (5 PROMPTS)

11. "Improve this paragraph but keep my voice: [paste]"
12. "Give me 3 source citations for [topic]."
13. "Write a strong opening for [essay type]."
14. "Strengthen this conclusion: [paste]"
15. "Fix grammar only. Do not change my style."

CAREER (5 PROMPTS)

16. "Write a GitHub README for a DS student."
17. "Write a LinkedIn summary for [skills] learner."
18. "Suggest 5 portfolio projects for beginners."
19. "Draft a cold email for [service] to [audience]."
20. "Review this resume bullet: [paste]. Improve it."



// QUICK-REFERENCE CHEAT SHEET

CORE Framework C: Context | O: Objective | R: Format | E: Example

Best Starter "Explain [topic] like I'm a first-year student. Use examples."

Debugging Template "I got [error]. My code: [paste]. Expected: [X]. Actual: [Y]."

Persona Activation "Act as a [ROLE]. I am [LEVEL]."

Chain-of-Thought Add: "Think step by step" for math/logic/debugging.

Refinement "Close, but fix [SPECIFIC]. Apply [QUALITY] to the rest."

Multi-Persona Same prompt → 2 personas → ask AI to compare answers.

Submission Check Ask yourself: level? objective? format? example?

Golden Rule Never accept the first answer for anything important.

Toolkit Rule Spend 6 hours Day 1 on 8 tools. Free tiers. Start now.

// ABOUT THE AUTHOR

Joy Roy is the founder of AI Blog Factory — a studio creating premium digital guides for students, creators, and tech learners.

A BSc Data Science student who built these systems in real courses with real deadlines. Every technique in this prompt engineering guide was tested, refined, and verified through hundreds of real AI conversations.

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